



Structure–property relationships of Fe₂O₃ doped novel oxyfluorophosphate glasses



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ABSTRACT

The relationship between the network structure and properties of iron containing novel oxyfluorophosphate glasses with the composition $20\text{ZnO}-20\text{CaF}_2-(60-x)\text{P}_2\text{O}_5:x\text{Fe}_2\text{O}_3$ with $0 \leq x \leq 5$ mol% has been studied. The density, molar volume, oxygen molar volume and oxygen packing density studies helped to understand the structural changes occurring in these glasses. The dissolution rate and pH of the immersion liquid are measured to explain the chemical durability of the glasses. Differential thermal analysis studies are carried out to analyze the thermal stability of the studied glasses. Deconvoluted infrared and Raman spectra of the glasses are analyzed to determine the relative areas of the vibrational bands corresponding to different structural units. Analysis of the vibrational spectra in conjunction with the other properties reveals that Fe₂O₃ enters the glass matrix in the form of Fe²⁺ and Fe³⁺ ions through the formation of more stable P–O–Fe²⁺ and P–O–Fe³⁺ bonds at the expense of easily hydrated P–O–P bonds.

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1. Introduction

Glasses exhibit a broad variety of properties which make them probable elements in the modern scientific and technological applications [1,2]. In the recent past, glasses are vital in the advancement of solid state lasers, optical glass fibers, biomaterials, imaging technologies, and glass films in microelectronic devices [3–7]. Phosphate glasses are of increasing interest nowadays because of their broad range of applications. Due to the low melting and softening temperatures, low viscosity and high thermal expansion coefficient [8], phosphate glasses possess multiple applications such as low temperature sealing materials [9], biomedical, quick ionic conductors [10], solid state electrolytes, glass-to-metal seals, optical data transmission, laser technology [11], thick film paste, low temperature enamels for metals [12] etc. However, the relatively poor chemical durability of phosphate glasses, limits them only to industrial applications. It has been proved that chemical durability of phosphate glasses can be enhanced by the addition of various modifier oxides such as ZnO, CaO, BaO, PbO, Li₂O, K₂O, and Na₂O [13–20].

Zinc oxide glasses are important due to their non-toxicity, non-hygroscopic nature, lower cost and useful optical, electrical and magnetic properties [21–23]. ZnO can act as both a six-coordinated network modifier and four-coordinated network former [24,25]. Zn²⁺ ions in oxide glasses enhance mechanical properties and chemical resistivity through the formation of tetrahedral co-ordination. ZnO is a key

constituent for the preparation of multicomponent oxide glasses with a high thermal resistance against crystallization [26,27]. It has been established that ZnO in oxide glasses can lower the melting point and glass transition temperature and improve the thermal expansion coefficient [28,29]. Zinc phosphate glasses find applications in low temperature seals and can be used as luminescent ion hosts. In recent times, it has been revealed that the optical properties of zinc phosphate glasses can be tailored with a femto-second laser. Hence, these glasses exhibit a variety of applications in making optical devices [30]. Fining agents such as calcium fluoride are added to the glass mixture to remove bubbles from the melt [2,31]. Important biological applications for calcium phosphate glasses also exist as it was demonstrated that they are biocompatible as bones and dental implants [32,33]. The addition of CaF₂ into the glass atmosphere lowers the viscosity liquidus temperature to a significant amount and further it acts as a good mineralizer. Also, fluorine ions coming from CaF₂ act as co-activators and smooth the progress of the substitution of activators into the lattice [34]. The structure and properties of oxide glasses containing fluorine ions are important owing to their potential applications in the fields of infrared fiber optics, laser windows and multifunctional optical components [2].

The addition of transition metal oxide into the glass causes a change in the glass configuration in which the metal oxide acts as a modifier or network former [35]. Extensive investigations on the optical absorption, luminescence and ESR spectroscopy of different transition metal ions in a variety of phosphate glasses have been made in the recent years in view of their technological applications [36–40]. Transition metal ions are known to influence the optical, electrical and magnetic properties

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of glasses due to their high sensitive response to the changes in the surrounding actions. Further, these ions can be used as a better candidate to probe the glass structure due to their broad radial distribution of outer *d*-orbital electron functions [41]. Transition metals present in-depth knowledge about many detailed aspects like the geometry of structural units present in the glass, the character of chemical bonds as well as the coordination polyhedra [42].

Iron ions have strong bearing on electrical, optical and magnetic properties of glasses. A large number of interesting studies are available on the environment of iron ion in various inorganic glass systems viz., silicate, borate, phosphate and germanate glasses [43–49]. These ions exist in different valence states with different coordinations in glass matrices, for example as Fe^{3+} with both tetrahedral and octahedral and as Fe^{2+} with octahedral environment [50,51]. The content of iron in different forms in different valence states existing in the glass depends on the quantitative properties of modifiers and glass formers, size of the ions in the glass structure, their field strength, mobility of the modifier cation etc. Hence, the connection between the state and the position of the iron ion and the physical properties of the glass is highly interesting. Further, it is also quite likely for iron ions to have a link with phosphate groups perhaps due to the creation of some $\text{Fe}-\text{O}-\text{P}$ bonds [52]; strengthen glass structure and may raise the chemical resistance of the glass.

Iron doped phosphate glasses are technologically as well as biologically important as they are used as biomaterials with modified rates of degradation in aqueous environment [53] and are used for the arrest of nuclear waste [54]. The addition of iron to phosphate glasses shows a momentous effect on the glass transition temperature and thermal expansion coefficient and enhances the chemical durability considerably. The addition of Fe_2O_3 into the glass environment causes a depolymerization of the phosphate chains complex. Also, iron phosphate glasses show corrosion-resistance, reinforcing fibers for composite materials [55]. Depending on the redox state and coordination number of Fe, iron phosphate glasses demonstrate fascinating electrical and magnetic properties [56–58].

Detailed literature survey shows that work has been carried out by researchers on the preparation, characterization, physical, vibrational and magnetic properties of phosphate glasses containing iron ions [59–63]. However, no effort appears to have been made to investigate the structure and properties of $\text{ZnO}-\text{CaF}_2-\text{P}_2\text{O}_5$ glasses doped with Fe_2O_3 .

The objective of the present work is to study the influence of Fe_2O_3 addition on the structure of $\text{ZnO}-\text{CaF}_2-\text{P}_2\text{O}_5$ glass and the mechanism of the conversion of different phosphate units in this glass with the aid of density, molar volume, infrared and Raman spectral studies. The importance of the present study is that it establishes the foundation for the structure of phosphate based glasses useful for industrial, biomedical and technological applications. Also, it is proposed to study the thermal parameters of the $\text{ZnO}-\text{CaF}_2-\text{P}_2\text{O}_5:\text{Fe}_2\text{O}_3$ glasses using differential thermal analysis which determines the qualitative estimation of the thermal stability and glass forming ability.

2. Experimental methods and characterization

2.1. Glass preparation

Glasses of the composition $20\text{ZnO}-20\text{CaF}_2-(60-x)\text{P}_2\text{O}_5:x\text{Fe}_2\text{O}_3$ with $0 \leq x \leq 5$ were prepared from appropriate amounts (all by mol%) of reagent grades of ZnO , CaF_2 , P_2O_5 and Fe_2O_3 ; among various compositions, this range seems to have formed a relatively clear and transparent glass. The powders of the raw materials were thoroughly mixed in an agate mortar and melted in a sealed silica crucible at 1173 K and kept in a microcontroller regulated pit furnace (Krishna Enterprises, Hyderabad, India) for about 30 min until a bubble free liquid was formed. The resultant melt was then poured in a graphite mold and subsequently annealed at 573 K for 24 h. The samples were then ground and optically polished. It may be noted here that, to fix the base or parent glass to which doping is performed, actually different combinations of $\text{ZnO}-\text{CaF}_2-\text{P}_2\text{O}_5$ glasses are prepared and among these combinations it is found that $20\text{ZnO}-20\text{CaF}_2$ glass sample is found to be more stable and chemically more durable and hence in the present study we preferred to this ratio of ZnO and CaF_2 in the parent glass.

2.2. Bulk glass characterization

2.2.1. ICP analysis

To analyze the final composition of the glasses, inductively coupled plasma atomic emission spectroscopy (Philips XL-30 spectrophotometer) was used. The nominal and analyzed glass compositions along with the codes of the glass samples under study are presented in Table 1.

2.2.2. Density and chemical durability measurements

The macroscopic density (ρ/kgm^{-3}) of the glasses at room temperature was determined to an accuracy of $\pm 4 \text{ kg m}^{-3}$ by standard principle of Archimedes using *o*-xylene (99.99% pure) as the buoyant liquid using the relation [39]:

$$\rho = \rho_b \frac{W_a}{W_a - W_b} \quad (1)$$

where ρ_b is the density of the buoyant liquid, and W_a and W_b are the weights of the samples in air and in the buoyant liquids respectively. The measurements were done with a digital balance (Shimadzu AUY 220 accuracy 0.0001 g) and the experiment was repeated a number of times for each composition and the average value was taken.

The chemical durability of the bulk glasses was determined from the average weight loss of the samples immersed in distilled water ($\text{pH} = 6.8 \pm 0.1$) for few minutes to 10 days. The glass samples of $1 \text{ cm} \times 1 \text{ cm} \times 0.6 \text{ cm}$ were cut, ground, polished to 600 grit finish with SiC paper, cleaned with acetone, weighed ($\pm 0.01 \text{ mg}$) and suspended by a weightless strand in polyethylene bottles filled with 100 ml of distilled water. The bottles were introduced in a constant temperature oven at

Table 1

Sample codes, nominal and analyzed batch compositions ($\pm 0.1\%$), average dissociation rate of $\text{ZnO}-\text{CaF}_2-\text{P}_2\text{O}_5:\text{Fe}_2\text{O}_3$ glasses and pH of the immersion liquid.

Sample code	Nominal (mol%)				Analyzed (mol%)				Avg diss rate ($\pm 10\%$) ($\text{D}_R/\text{g cm}^{-2} \text{ min}^{-1}$)	pH (± 0.1)
	ZnO	CaF ₂	P ₂ O ₅	Fe ₂ O ₃	ZnO	CaF ₂	P ₂ O ₅	Fe ₂ O ₃		
ZCP	20	20	60	0	19.8	19.6	60.6	0	3.4×10^{-7}	5.4
ZCPFe 0.5	20	20	59.5	0.5	20.1	19.9	59.6	0.4	5.5×10^{-7}	5.3
ZCPFe 1.0	20	20	59	1.0	19.8	20.2	58.9	1.1	7.2×10^{-7}	5.0
ZCPFe 1.5	20	20	58.5	1.5	19.9	20	58.5	1.6	1.0×10^{-6}	4.8
ZCPFe 2.0	20	20	58	2.0	20	19.7	58.4	1.9	5.4×10^{-7}	5.2
ZCPFe 3.0	20	20	57	3.0	19.9	19.9	57.1	3.1	3.1×10^{-8}	5.6
ZCPFe 4.0	20	20	56	4.0	20.3	19.8	56	3.9	9.1×10^{-9}	5.8
ZCPFe 5.0	20	20	55	5.0	20.2	20.1	54.9	4.8	2.5×10^{-9}	6.1

363 ± 1 K. The weight loss of the samples was measured at regular time intervals. The error in weight loss was estimated at ± 10%. The dissolution rate ($D_R/\text{g cm}^{-2} \text{ min}^{-1}$), normalized to the glass surface area (S/cm^2) and the immersion time (t/min), was calculated from the weight loss ($\Delta W/\text{g}$) of the sample measured at various times using the relation [59]:

$$D_R = \frac{\Delta W}{St} \quad (2)$$

For aqueous durability, all tests were made in duplicate and the average value of D_R is reported. Also, the pH of the water was measured at regular time intervals using a digital pH meter. The average dissociation rate of the glass samples and the pH value of the immersion liquid are presented in Table 1.

2.2.3. Differential thermal analysis recordings

The differential thermal analysis of the samples in the temperature range 303–1173 K was carried out on a Perkin-Elmer DTA-7 by heating 35–40 mg of glass powders with a programmed heating rate of 10 K/min in high purity open alumina crucibles under nitrogen flow. The characteristic temperatures useful for explaining the thermal stability of the glasses were determined with an estimated uncertainty of ± 1 K.

2.3. Structural characterization

2.3.1. X-ray diffraction spectral patterns

The X-ray diffraction spectra of the glasses under investigation were recorded on Bruker D8 diffractometer in the step scan mode, with $\text{Cu K}\alpha$ radiation of wavelength 1.542 Å at a voltage of 30 kV and 20 mA anode current with a step value of 0.02° per 3 s from 0° to 90°. The absence of peaks corresponding to different crystalline phases in the spectra confirmed the amorphous state of the glasses.

2.3.2. Vibrational spectroscopic techniques

The infrared spectra of the glass samples were recorded at room temperature in the wavenumber range 1500–400 cm^{-1} with a resolution of 1 cm^{-1} by an IR spectrophotometer of type alpha by Bruker powered by a 240 V and 10 A ac supply using the KBr pellet technique. The resulting spectra were deconvoluted and peak fitted for a comprehensive study of the glass structure. DeltaNu Advantage NIR Raman spectrometer was used to record the Raman spectra of the glass samples in the wavenumber range 1400–200 cm^{-1} with a resolution of 1 cm^{-1} .

3. Results

3.1. Density and molar volume

Density is considered as an important physical property that can easily detect minute structural changes in the glass network [64]. From the measured densities (ρ) at room temperature and average

molar molecular weights ($\bar{M}$), the physical parameters like molar volume (V_m), oxygen molar volume (V_o), oxygen packing density (OPD), transition metal ion concentration (N_i), interionic distance (r_i), polaron radius (r_p) and field strength (F) useful for understanding the structural changes taking place in $\text{ZnO-CaF}_2\text{-P}_2\text{O}_5$ glasses with the addition of Fe_2O_3 dopant are computed using the relations mentioned in the literature [65,66] and are presented in Table 2. The variation of density, molar volume, oxygen molar volume and oxygen packing density of ZCP glasses with the mol% of Fe_2O_3 is pictorially represented in Fig. 1. From the figure, it is found that the density and oxygen packing density of these glasses first decrease up to 1.5 mol% and then increase with the content of Fe_2O_3 ; in contrast the molar volume and oxygen molar volume showed an exactly opposite trend.

3.2. Chemical durability

The variation of average dissolution rate of the glass samples and the pH value of the liquid in which the samples are immersed with the mol% of Fe_2O_3 is shown in Fig. 2. It is clear from the figure that, the average dissociation rate (D_R) increases from $3.4 \times 10^{-7} \text{ g cm}^{-2} \text{ min}^{-1}$ for pure glass (ZCP) to $1.0 \times 10^{-6} \text{ g cm}^{-2} \text{ min}^{-1}$ for 1.5 mol% of Fe_2O_3 doped glass and then drastically decreases to 2.5×10^{-9} for 5 mol% of Fe_2O_3 doped glass. It is also noticed that the pH value of the immersion liquid followed an opposite trend with highest value of 6.1 (Table 1) for the glass with highest mol% of Fe_2O_3 (ZCPFe 5.0).

3.3. Differential thermal analysis

Differential thermal analysis is a technique for obtaining the glass characteristic temperatures, such as the glass transition temperature, T_g (the midpoint of the change in heat capacity in the glass transition region), crystallization temperature, T_c (the onset of the crystallization peak), peak temperature, T_p (the maximum of the crystallization peak) and melting temperature, T_m (onset of the melting) which are useful in estimating the glass forming ability and the thermal stability of the glasses [67]. Typical differential thermal analysis curves of glass samples ZCP, ZCPFe 0.5, ZCPFe 1.5 and ZCPFe 5.0 in the temperature range 600–1173 K are presented in Fig. 3 and the characteristic temperatures obtained from DTA curves are reported in Table 3. From the table, it is noticed that the glass transition temperature (T_g) decreases from 694.3 K (pure) to 683.1 K (1.5 mol% of Fe_2O_3) and then increases to 704.8 K (5 mol%). It is also observed that the other three characteristic temperatures (T_c , T_p and T_m) followed the same trend as T_g with the change of Fe_2O_3 content in the glasses. The graphical variation of the glass transition temperature with the mol% of Fe_2O_3 is shown in Fig. 4.

3.4. Raman spectra

Fig. 5 represents the Raman spectra of $\text{ZnO-CaF}_2\text{-P}_2\text{O}_5\text{:Fe}_2\text{O}_3$ glasses in the wavenumber region 200–1400 cm^{-1} recorded at room temperature. The pure glass (ZCP) has exhibited Raman peaks characteristic of vibrational bands of phosphorous-oxygen and metal cation linkages

Table 2
Density and related physical properties of Fe_2O_3 doped oxyfluorophosphate glasses.

Glass	ρ (kgm^{-3})	$\bar{M}$ (kg-mol^{-1})	V_m ($10^{-6} \text{ m}^3 \text{ mol}^{-1}$)	V_o ($10^{-6} \text{ m}^3 \text{ mol}^{-1}$)	OPD (10^3 mol m^{-3})	N_i ($10^{26} \text{ ions m}^{-3}$)	r_i (Å)	r_p (Å)	F ($\text{\AA}^{-2}$)
ZCP	2754	117.063	42.51	13.28	75.28	–	–	–	–
ZCPFe 0.5	2731	117.152	42.90	13.45	74.36	70.2	5.22	2.10	0.68
ZCPFe 1.0	2695	117.241	43.50	13.68	73.10	138.4	4.16	1.68	1.07
ZCPFe 1.5	2668	117.330	43.98	13.87	72.08	205.4	3.65	1.47	1.39
ZCPFe 2.0	2681	117.418	43.80	13.86	72.15	275.0	3.31	1.34	1.68
ZCPFe 3.0	2722	117.596	43.20	13.76	72.68	418.2	2.88	1.16	2.23
ZCPFe 4.0	2771	117.773	42.50	13.62	73.41	566.8	2.60	1.05	2.73
ZCPFe 5.0	2839	117.951	41.55	13.40	74.62	724.9	2.40	0.97	3.21
Error (–)	±4	±0.001	±0.04	±0.02	±0.12	±0.2	±0.4	±0.4	±0.2

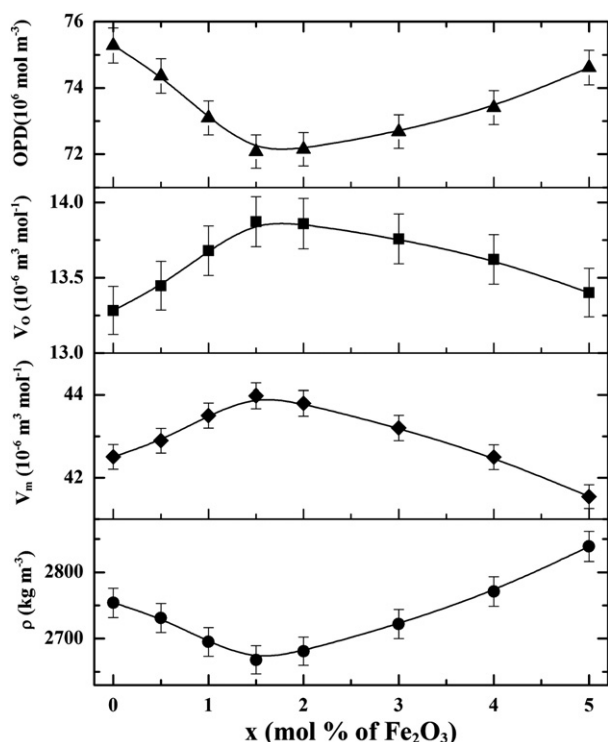


Fig. 1. Variation of density (ρ), molar volume (V_m), oxygen molar volume (V_o) and oxygen packing density (OPD) of ZnO–CaF₂–P₂O₅ glasses with the mol% of Fe₂O₃ (lines are drawn as a guide to the eye).

corresponding to different structural units. The low intensity peaks observed at 264, 342 and 687 cm⁻¹ in the frequency region 200–800 cm⁻¹ of the Raman spectrum of pure glass can be easily identified and assigned; however, the spectrum in the wavenumber range 800–1400 cm⁻¹ is complicated with a number of indistinguishable peaks and shoulders. The peaks at 264 and 342 cm⁻¹ are respectively assigned to ZnO₄ structural units [1] and bending of (PO₄) units with cation as modifier [68] and the relatively more intense peak at 687 cm⁻¹ is attributed to the symmetric stretching vibrations of the bridging oxygens of P–O–P linkages of (PO₃)²⁻ groups in Q¹ structural units [69]. In the Fe₂O₃ doped glasses an additional peak due to Fe–O bonds [68] is observed between 508 and 535 cm⁻¹ in the low wavenumber region. As most of the peaks at high frequency (800–1400 cm⁻¹) are large and asymmetric, presenting also some shoulders as observed and to exactly locate the peak positions with the relative area under each

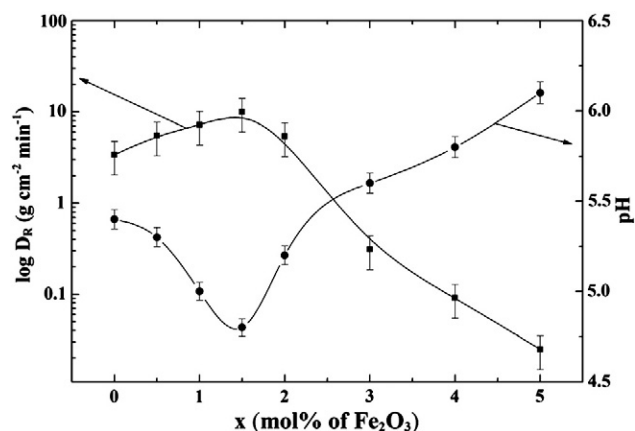


Fig. 2. Change in the dissolution rate (D_R) of ZnO–CaF₂–P₂O₅ glasses and pH of the immersion liquid with the mol% of Fe₂O₃ (lines are drawn as a guide to the eye).

peak, a deconvolution of the experimental spectra was necessary. The deconvolution process and the analysis of the bands have been described and discussed elsewhere [70,71]. As an example, the deconvoluted Raman spectra of the pure (ZCP) and 1.5 mol% of Fe₂O₃ doped (ZCPFe 1.5) glasses are shown in Fig. 6. The assignment of the vibrational bands obtained in Raman spectra of Fe₂O₃ doped ZnO–CaF₂–P₂O₅ glasses is presented in Table 4 and the deconvolution data (band center and relative areas under bands) of all the investigated glasses are given in Table 5. The variation of the total relative areas (symmetrical and asymmetrical stretching vibrations) of different structural units obtained after the deconvolution of Raman spectra of ZCP glasses with the mol% of Fe₂O₃ is shown in Fig. 7.

3.5. Fourier transform infrared spectra

Normalized FTIR absorbance spectra of ZnO–CaF₂–P₂O₅:Fe₂O₃ glasses in the wavenumber region 1600–400 cm⁻¹ at room temperature are shown in Fig. 8. Experimental IR spectrum of pure ZnO–CaF₂–P₂O₅ glass (ZCP) has exhibited vibrational bands characteristic of phosphate and metal cation structural units which can be divided into three regions with complex bands: (i) wavenumbers between 1400 and 1120 cm⁻¹ as region of the stretching vibrations of P=O bonds of Q³ tetrahedra and (PO₂)⁻ groups of Q² tetrahedra with the non-bridging oxygen atoms (NBO), (ii) between 1120 and 750 cm⁻¹ as region of the stretching vibrations of (PO₄)³⁻ groups of Q⁰ units and P–O–P linkages of Q¹ tetrahedra and (iii) region below 600 cm⁻¹ as region of all deformation and network vibrations of the cations. The addition of Fe₂O₃ to the glass has produced considerable changes in the position and intensity of the vibrational bands in these regions of the spectra indicating the structural readjustment of the glass network.

From the normalized infrared spectra one can obtain only qualitative information. This explanation gives no account for the specific phosphate groups containing different structural units. A comprehensive elucidation of the infrared spectra of the glasses containing P₂O₅ is not an easy task due to the intricacy originating from the glassy nature of the materials and to the large number of possible structural groups found in phosphate glasses. However, each of the aforesaid absorption bands clearly exhibits fine structure and thus a more detailed interpretation can be attempted.

The IR absorption spectra of the studied glasses appear to be somewhat complicated due to their overlapping bands or nearby positions and from the first sight appear to be more or less similar for all the glasses under study. To realize and confirm that the network structure of ZnO–CaF₂–P₂O₅ glass is affected by the addition of Fe₂O₃ and to get more information about the structural groups in the glasses, which may be sensitive to the intermediate range order changes in the glass matrix, a quantitative analysis is carried out by the deconvolution of IR spectrum which also provides the mechanism of conversion of various phosphate structural units. The deconvolution process and the analysis of the bands have been described and discussed elsewhere. The process makes it possible to calculate the relative area of each component peak [70–72]. Each component peak is related to some type of vibration in a specific structural unit. The concentration of the structural unit is proportional to the relative area of its component band. A typical deconvolution in Gaussian bands of IR spectra of the pure (ZCP) and 5 mol% of Fe₂O₃ doped (ZCPFe 5.0) glasses is illustrated in Fig. 9. It is found that the difference between experimental and simulated curves is less than 0.15%. The deconvolution parameters viz., the band center C and the relative area A for the investigated glasses are given in Table 6. It is found to be useful to restrict the deconvolution treatment to the spectrum part in the range 400 to 1600 cm⁻¹ which represents the vibrations of the main building units of the phosphate network along with metal cation structural units. The wavenumber assignment of various peaks observed in the deconvoluted FTIR spectra to different phosphate and other structural units is given in Table 4.

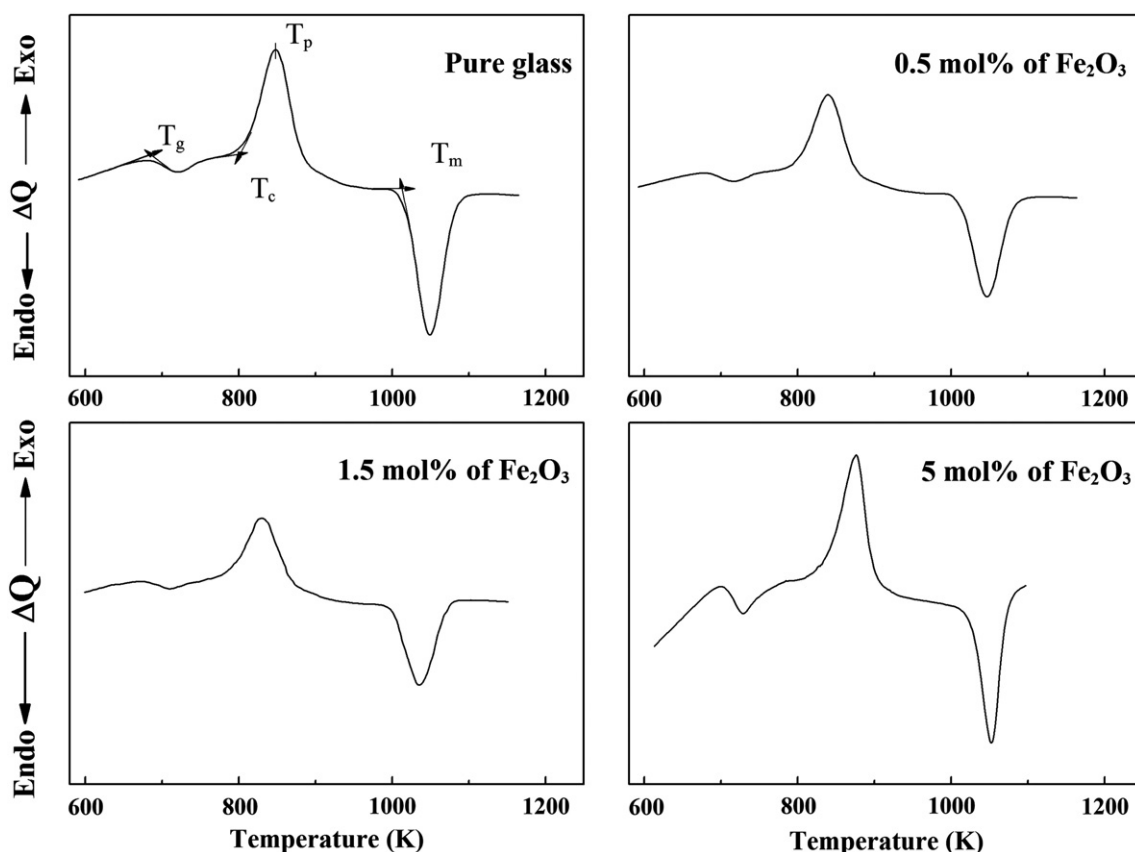


Fig. 3. Differential thermal analysis curves of ZnO–CaF₂–P₂O₅:Fe₂O₃ glasses.

4. Discussion

Vitreous P₂O₅ consists of Q³ type PO₄ tetrahedra in which three of the oxygens are bridging (P–O–P) and one is terminal oxygen (P=O). Addition of modifiers like ZnO and CaF₂ to P₂O₅ converts three-dimensional network into linear phosphate chains and end groups with the cleavage of P–O–P linkages and creation of non-bridging oxygen atoms in the glass which has an intrinsic polymeric feature [3]. P–O–P bridging oxygens are converted to P–O–M⁺ non-bridging oxygens (where M⁺ = Zn²⁺ or Ca²⁺), which indicates the formation of Q² tetrahedra with two bridging oxygen atoms, Q¹ tetrahedra with one bridging oxygen atom, and Q⁰ tetrahedra with no bridging oxygen atoms [3].

The addition of the dopant Fe₂O₃ to ZnO–CaF₂–P₂O₅ glass further replaces some of the easily hydrated P–O–P bonds and the P–O–Zn/Ca bonds by moisture and corrosion resistant chemically more stable P–O–Fe linkages [54,60]. This replacement does not result in the decrease of Zn/Ca ions but through the conversion/formation of MO_{4/6} (M = Zn, Ca). It may be noted here that in binary phosphate glasses

only two types of the 4 polyhedra exist [9,73] whereas in multicomponent phosphate glasses, all or some of the four polyhedra exist [68,74]. These linkages and the structure of the glasses are well reflected in the properties discussed below.

4.1. Density and molar volume studies

In general, the structural compactness, modification of the geometrical configuration of the glassy network, change in the coordination of the glass forming ions and fluctuations in the dimensions of the interstitial holes are the factors that influence the density of a glass [75]. The observed behavior in the density and molar volume values (Fig. 1) suggests some sort of rearrangement taking place in the network of these glass samples [76]. But, from XRD and IR spectra, it is obvious

Table 3

Glass characteristic temperatures and various thermal stability factors of Fe₂O₃ doped ZCP glasses (characteristic temperatures are recorded with an accuracy of ±1 K).

x (mol%)	T _g (K)	T _c (K)	T _p (K)	T _m (K)	ΔT (K)	S (K)	H _r
0	694.3	800.8	851.3	1003.1	106.5	7.746	0.526
0.5	689.9	793.1	842.7	999.4	103.2	7.420	0.500
1.0	686.7	786.6	835.8	996.6	99.9	7.158	0.476
1.5	683.1	779.7	829.0	992.3	96.6	6.972	0.454
2.0	686.3	785.7	835.1	996.9	99.4	7.155	0.471
3.0	693.2	799.3	848.9	1005.4	106.1	7.592	0.515
4.0	700.6	814.3	863.9	1013.4	113.7	8.050	0.571
5.0	704.8	825.8	875.3	1017.1	121.0	8.498	0.633

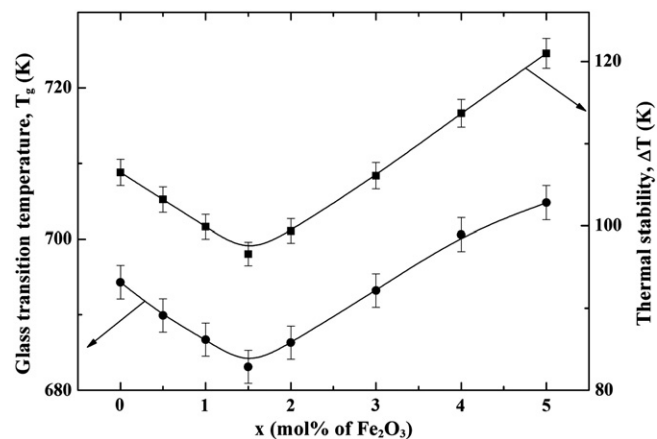


Fig. 4. The variations in glass transition temperature (T_g) and thermal stability (ΔT) of oxyfluorophosphate glasses with the mol% of Fe₂O₃ (lines are drawn as a guide to the eye).

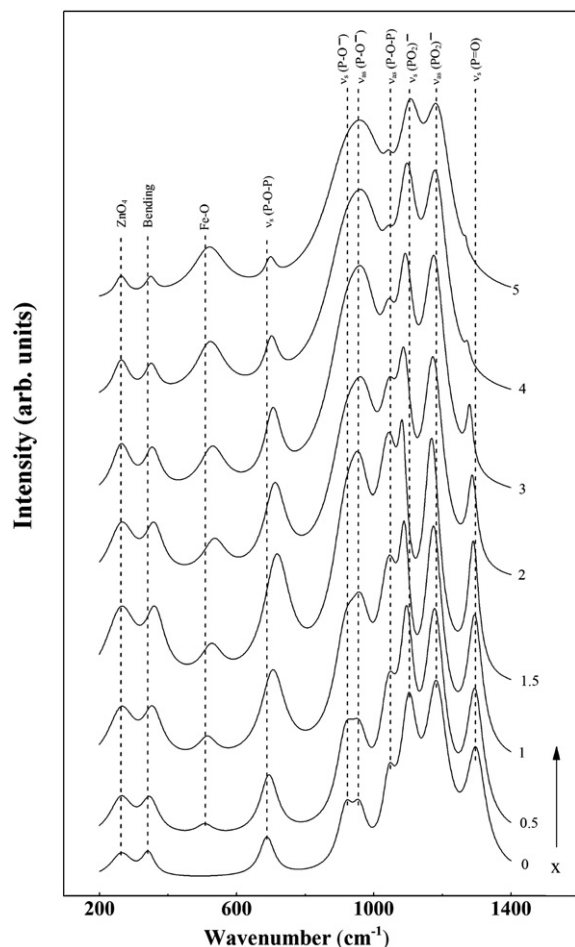


Fig. 5. Raman spectra of ZnO-CaF₂-P₂O₅:Fe₂O₃ glasses.

that these glass samples are still in an amorphous state indicating that there is no detectable change to crystallization. Hence, the variation in density can be attributed only to the differences in linkages between different structural species.

The decrease in density and the increase in molar volume (Fig. 1) up to 1.5 mol% of Fe₂O₃ are unexpected as Fe₂O₃ has a higher density than P₂O₅. This decrease in density is due to the expansion of Q³ network to accommodate Fe²⁺ and Fe³⁺ ions in the interstices of PO₄ network indicating the formation of P–O–Fe²⁺ and P–O–Fe³⁺ bonds [57,63]. Hence, a conversion of fraction of Q³ and Q² units to Q¹ and Q⁰ units may be expected up to 1.5 mol% of Fe₂O₃. Also, the decrease in the density up to 1.5 mol% of Fe₂O₃ can be attributed to the increase in the concentration of FeO₆ octahedral units in the network. The increase in the density above 1.5 mol% of Fe₂O₃ may be due to the appearance of more corner and edge sharing metal polyhedra in the glass network indicating the conversion of Q¹ units to Q² and Q⁰ units in accordance with the disproportionation equation $2Q^1 \leftrightarrow Q^2 + Q^0$ [9,30].

Also, iron ions in the glass network exist usually in both four (Fe³⁺) and six (Fe²⁺, Fe³⁺) fold coordinations, whereas P⁵⁺ ion in the glass network is usually in fourfold coordination. Therefore, the average coordination number of the cation in the glass network increases with increasing Fe₂O₃ content above 1.5 mol%, which improves glass compactness [59]. Hence, an increase in the density can be expected above 1.5 mol% of Fe₂O₃. Thus, the increase in the density above 1.5 mol% of Fe₂O₃ may also be due to the formation of FeO₄ tetrahedral structural units through some Fe–O–P bonds which probably results in better packing causing strengthening of the glass network.

The observed increase in the oxygen molar volume (V_o) from 13.28×10^{-6} to $13.87 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}$ with the increase in the content of Fe₂O₃ up to 1.5 mol% indicates the loose packing of the atoms in the glass structure. Further increase in the content of Fe₂O₃ showed a decrease in oxygen molar volume to $13.40 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}$ which indicates the increased packing of atoms in the glass network structure [1]. These results for oxygen molar volume (V_o) and oxygen packing density (OPD) confirm a change in the topology of the glass

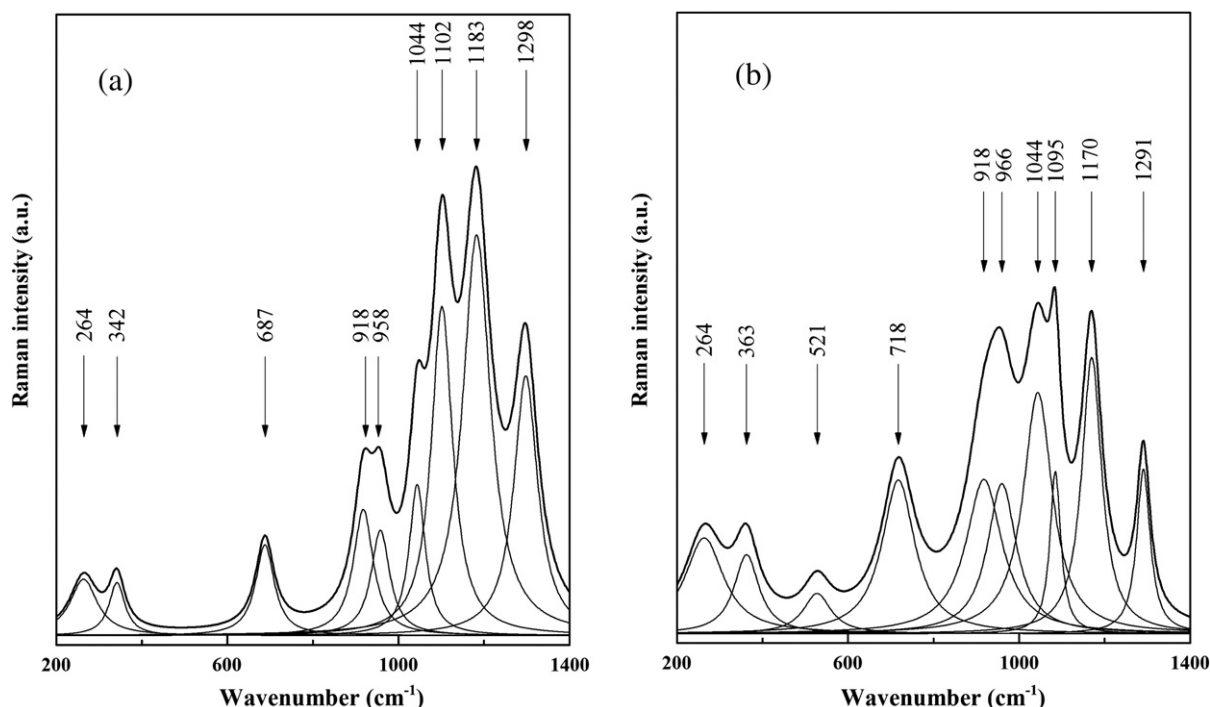


Fig. 6. Deconvoluted Raman spectra of (a) $x = 0$ mol% and (b) $x = 1.5$ mol% of Fe₂O₃ containing oxyfluorophosphate glasses.

Table 4Assignments of Raman and FTIR bands of Fe₂O₃ doped ZCP glasses (wavenumbers are recorded with an accuracy of $\pm 1 \text{ cm}^{-1}$).

Wavenumber (ν/cm^{-1})		Assignment
Raman	Infrared	
264	460	Specific vibrations of Zn–O bonds in ZnO ₄ units
342–363	513–528	Network deformation modes
508–535	–	Fe–O bonds
–	564–581	Octahedral FeO ₆ units
–	633–659	Tetrahedral FeO ₄ units
687–718	777	Symmetric stretching vibrations of the bridging oxygens of P–O–P linkages of (PO ₃) ^{2–} groups in Q ¹ structural units
1044	893–925	Asymmetric stretching vibrations of the bridging oxygens of P–O–P linkages of (PO ₃) ^{2–} groups in Q ¹ structural units
918	978–997	Symmetric stretching vibrations of P–O [–] bonds of (PO ₄) ^{3–} groups in Q ⁰ units
958–979	1086–1106	Asymmetric stretching vibrations of P–O [–] bonds of (PO ₄) ^{3–} groups in Q ⁰ units
1085–1106	1139–1160	Symmetric stretching vibrations of non-bridging oxygens of O–P–O linkages of (PO ₂) [–] groups in Q ² units
1170–1187	1220–1252	Asymmetric stretching vibrations of non-bridging oxygens of O–P–O linkages of (PO ₂) [–] groups in Q ² units
1268–1298	1290	Symmetric vibrations of P=O bonds in Q ³ units
–	1368–1395	Asymmetric vibrations of P=O bonds in Q ³ units

structure with composition through the conversion of different structural units.

4.2. Chemical durability studies

It is well known that easily hydrated P–O–P bridging links which are predominant in the structure of common phosphate glasses are generally the reason for their poor chemical durability [59]. The addition of Fe₂O₃ has diverse effects on chemical durability of phosphate glasses; in some conditions it can reduce the durability of phosphate glasses if it enters the phosphate network as a modifier (FeO₆) and on the other hand, in some cases it could be responsible for the formation of phosphate-oxygen-iron linkages which generally improve the chemical durability if Fe₂O₃ enters the glass matrix as a network former (FeO₄). The observed increase in the average dissociation rate (Fig. 2) of the glasses with the increase in the content of Fe₂O₃ up to 1.5 mol% shows that the durability of the glass initially decreases indicating the increase in the concentration of FeO₆ octahedral units. Also, up to 1.5 mol% of Fe₂O₃, the cross-linking between different ions is less. The improvement in chemical durability of the glasses with increasing Fe₂O₃ content from 2 to 5 mol% can be attributed to the replacement of the easily hydrated P–O–P bonds by mechanically stronger and chemically more durable P–O–Fe bonds. The decreased dissociation rate above 1.5 mol% of Fe₂O₃ suggests the conversion of a fraction of Q¹ units into Q² and Q⁰ units. However, the mechanism and rate of conversion cannot be satisfactorily explained by variations in dissolution rates. Clearly, the glass containing 5 mol% of Fe₂O₃ possesses excellent aqueous durability comparable to window glass and can also meet the need for sealing applications [59].

The reduction in pH value of the immersion solution up to 1.5 mol% of Fe₂O₃ is due to the dissolution of phosphate species from these glasses and the subsequent formation of phosphoric acid [77]. This decrease in pH is consistent with the larger D_R values (Table 1). The increase in pH of the immersion liquid above 1.5 mol% of Fe₂O₃ is due to the increase of non-bridging oxygens that create cross-links between the phosphate chains which block the entrance of water molecules in to the glass network [78].

4.3. Thermal stability studies

The observed variation in the glass transition temperature (Fig. 4) with the increase in mol% of Fe₂O₃ suggests that some structural modifications are taking place in the glass network. The initial decrease in the glass transition temperature may be attributed to the loose packing of the atoms in the glass [79] due to the depolymerization of the network indicating the increased modifier action of iron ions in the glass network up to 1.5 mol% of Fe₂O₃. The increase in the glass transition temperature of the samples above 1.5 mol% of Fe₂O₃ may be due to the replacement of easily hydrated P–O–P bonds by corrosion and moisture resistant highly stable P–O–Fe bonds which enhance the cross-linking of the network due to the network forming role of iron ions [79]. The glass transition temperature studies indicate that the glass with 5 mol% of Fe₂O₃ is the most stable of all the samples under study.

In order to explain the thermal stability of the glasses in a more qualitative fashion, various parameters such as Dietzel's [80] thermal stability, $\Delta T = (T_c - T_g)$, Hruby's [81] glass forming ability factor, $H_r = \Delta T / (T_m - T_c)$ and Saad-Poulin's [82] thermal stability parameter $S = (T_p - T_c) / \Delta T$ are determined from the glass characteristic temperatures obtained from the differential thermal analysis curves of

Table 5Wavenumbers (C) and relative areas (A) of deconvoluted bands in the Raman spectrum of Fe₂O₃ doped oxyfluorophosphate glasses (accuracy of wavenumbers of band centers is $\pm 1 \text{ cm}^{-1}$).

x (mol%)		$\nu_s(\text{P}=\text{O})$	$\nu_{as}(\text{O}-\text{P}-\text{O})$	$\nu_s(\text{O}-\text{P}-\text{O})$	$\nu_{as}(\text{P}-\text{O}-\text{P})$	$\nu_{as}(\text{P}-\text{O}^-)$	$\nu_s(\text{P}-\text{O}^-)$	$\nu_s(\text{P}-\text{O}-\text{P})$	Fe–O	$\delta(\text{O}=\text{P}-\text{O})$	ZnO ₄
0	C	1298	1183	1102	1044	958	918	687	–	342	264
	A	17.5	31.5	19.8	6.9	5.6	7.1	4.8	–	2.4	4.4
0.5	C	1296	1179	1097	1044	959	918	695	508	348	264
	A	13.8	25.5	14.4	12.3	7.8	8.8	6.7	1.1	3.8	5.8
1	C	1295	1175	1090	1044	963	918	706	515	356	264
	A	10	19.4	7.9	16.2	9.9	11.7	10.8	2	4.8	7.3
1.5	C	1291	1170	1085	1044	966	918	718	521	363	264
	A	5.8	14.4	4.6	18.3	11.4	14.6	13.2	2.9	5.2	9.6
2	C	1287	1173	1088	1044	970	918	711	524	359	264
	A	3.9	19.1	9.5	8.9	15.3	18.5	9.1	4.2	4.3	7.2
3	C	1280	1177	1093	1044	973	918	706	528	355	264
	A	1.8	21.5	11.4	4.6	19.7	22.4	5.5	5.4	3.2	4.5
4	C	1274	1182	1098	1044	978	918	702	530	352	264
	A	0.3	21.8	14.5	1.8	21.5	24.7	2.8	7.6	1.9	3.1
5	C	1268	1187	1106	1044	979	918	699	535	350	264
	A	0.1	23.4	15.8	0.9	22.1	25.2	1.2	8.9	1	1.4

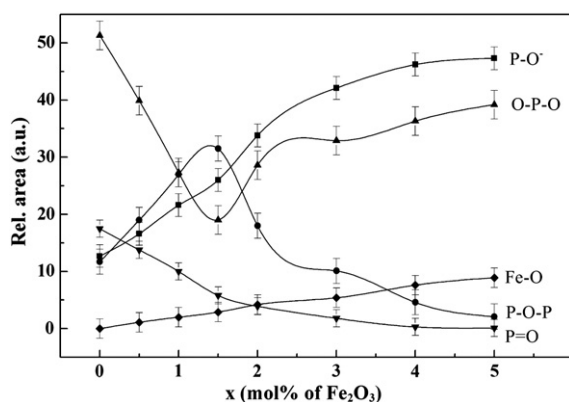


Fig. 7. Variation of the relative areas of Raman bands of different structural units in ZCP glasses with the mol% of Fe_2O_3 (lines are drawn as a guide to the eye).

the glasses and are presented in Table 3. These parameters predict the glass-forming ability of a material. The larger values of these factors signify the greater thermal stability of the glasses. The variation of thermal stability (ΔT) of ZCP glasses with the mol% of Fe_2O_3 is represented in Fig. 4. Thermal stability is very important for sealing applications, during which crystallization of vitreous frits during cooling or reheat treatment can affect their sealing process. A glass composition with higher values of ΔT (> 100 K) usually represent a higher thermal stability and glass-forming affinity [67]. From Table 3, thermal stability ΔT is found to be 96.6 K (lowest) at 1.5 mol% of Fe_2O_3 doped glass and 121 K (highest) for 5 mol% of Fe_2O_3 doped glass. The Hruby parameter (H_r) which represents the stability of glass against crystallization and the

thermal stability parameter (S) which reflects the opposition to devitrification after formation of the glass are observed to decrease up to 1.5 mol% of Fe_2O_3 and then increase above this content of Fe_2O_3 indicating the increased stability of the glasses when Fe_2O_3 content increases from 1.5 to 5 mol% [83].

From the trends observed in these three stability parameters (Table 3), it can be concluded that the stability of the glass is more when Fe_2O_3 content in the glass is above 1.5 mol% and the glass with 5 mol% of Fe_2O_3 is the most stable one.

4.4. Raman spectral studies

The deconvoluted Raman spectrum (Fig. 6a) in the high frequency region $800\text{--}1400\text{ cm}^{-1}$ of iron-free glass (ZCP) has exhibited six vibrational bands at wavenumbers 918, 958, 1044, 1102, 1183 and 1298 cm^{-1} . The bands at 918 and 958 cm^{-1} are identified respectively due to the symmetric and asymmetric stretching vibrations of $(\text{PO}_4)^{3-}$ groups in Q^0 units [74,84]; the band at 1044 cm^{-1} is assigned to asymmetric stretching vibrations of P–O–P linkages of $(\text{PO}_3)^{2-}$ groups in Q^1 structural units [84]. Two more intense vibrational bands obtained at 1102 and 1183 cm^{-1} respectively are attributed to the non-bridging oxygens of O–P–O linkages of $(\text{PO}_2)^-$ symmetric and asymmetric vibrations in Q^2 units [84,85]; the band at 1298 cm^{-1} is identified due to symmetric vibrations of P=O bonds in Q^3 units [84]. An extra vibrational band obtained at 508 cm^{-1} with the addition of 0.5 mol% of Fe_2O_3 is assigned to Fe–O bonds [68]. Considerable changes in the wavenumber and relative area of the peaks are evident with the increase in Fe_2O_3 in the glasses indicating that the structure of $\text{ZnO-CaF}_2\text{-P}_2\text{O}_5$ glass is affected by the doping of Fe_2O_3 . The relative areas of different Raman peaks are plotted as a function of the Fe_2O_3 content in Fig. 7. With the increase in Fe_2O_3 content from 0.5 to 5 mol%, it is observed that (i) the frequencies of the bands at 264 (ZnO_4), 918 (Q^0_s) and 1044 cm^{-1} (Q^1_{as}) remain constant, (ii) the wavenumbers of the bands at 342 (network deformation modes) and 687 cm^{-1} (Q^1_s) first increase up to 1.5 mol% of Fe_2O_3 and then decreases with a similar trend in relative areas, (iii) the wavenumbers and relative areas of the bands at 1102 (Q^2_s) and 1183 cm^{-1} (Q^2_{as}) due to non-bridging oxygens of O–P–O linkages of $(\text{PO}_2)^-$ groups followed a trend opposite to that of 687 cm^{-1} band (Q^1_s), (iv) the bands at 958 (Q^0_{as}) and 508 cm^{-1} (Fe–O) show a gradual increase both in relative area and wavenumber and (v) the band due to P=O bonds at 1298 cm^{-1} (Q^3_s) displays a continuous decrease in wavenumber as well as relative area.

Up to 1.5 mol% of Fe_2O_3 in $\text{ZnO-CaF}_2\text{-P}_2\text{O}_5$ glass, the increase in the relative areas of the vibrational bands due to ZnO_4 , bending modes, Q^1_s , Q^1_{as} , Q^0_s and Q^0_{as} units at the expense of Q^2_s , Q^2_{as} and Q^3_s bands indicates that Q^1 and Q^0 structural units are increasing with a concomitant decrease of Q^2 and Q^3 units.

With the further increase of Fe_2O_3 from 1.5 to 5 mol%, the decrease in intensity of the vibrational band at 264 cm^{-1} indicates the decrease of ZnO_4 units probably due to the formation of relatively more stable Fe–O bonds. Also, the decrease in both wavenumber and relative area of the band due to deformation modes indicates the strengthening of glass against devitrification up to 5 mol% of Fe_2O_3 . The observed decrease in relative area of the peaks at 687 and 1044 cm^{-1} shows that the bond strength of P–O–P linkages decreases leading to the reduction of the pyrophosphate (Q^1) units in the glass, as the concentration of Fe_2O_3 increases. It is also evident from the studies that the P=O bonds due to Q^3 units decrease as can be seen from the near disappearance of the band at 1298 cm^{-1} at 5 mol% of Fe_2O_3 content in the glass. The decrease in relative areas of all these bands is compensated by the increase in the relative areas of the bands at 918 and 958 cm^{-1} corresponding to orthophosphate (Q^0) units and the bands at 1102 and 1183 cm^{-1} corresponding to metaphosphate (Q^2) units, as reflected in Fig. 7. The excess Q^0 and Q^2 units in the glasses with $1.5 < x \leq 5$ result from the disproportionation of pyrophosphate Q^1 units according to the reaction, $2\text{Q}^1 \leftrightarrow \text{Q}^0 + \text{Q}^2$. This increase in Q^0 and Q^2 units from 1.5 to

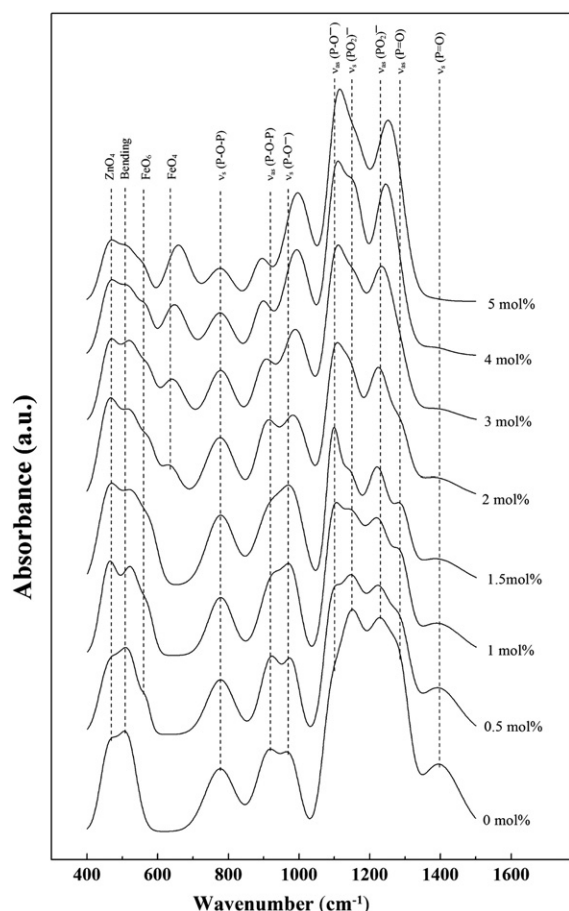


Fig. 8. Normalized FTIR absorbance spectra of Fe_2O_3 doped oxyfluorophosphate glasses.

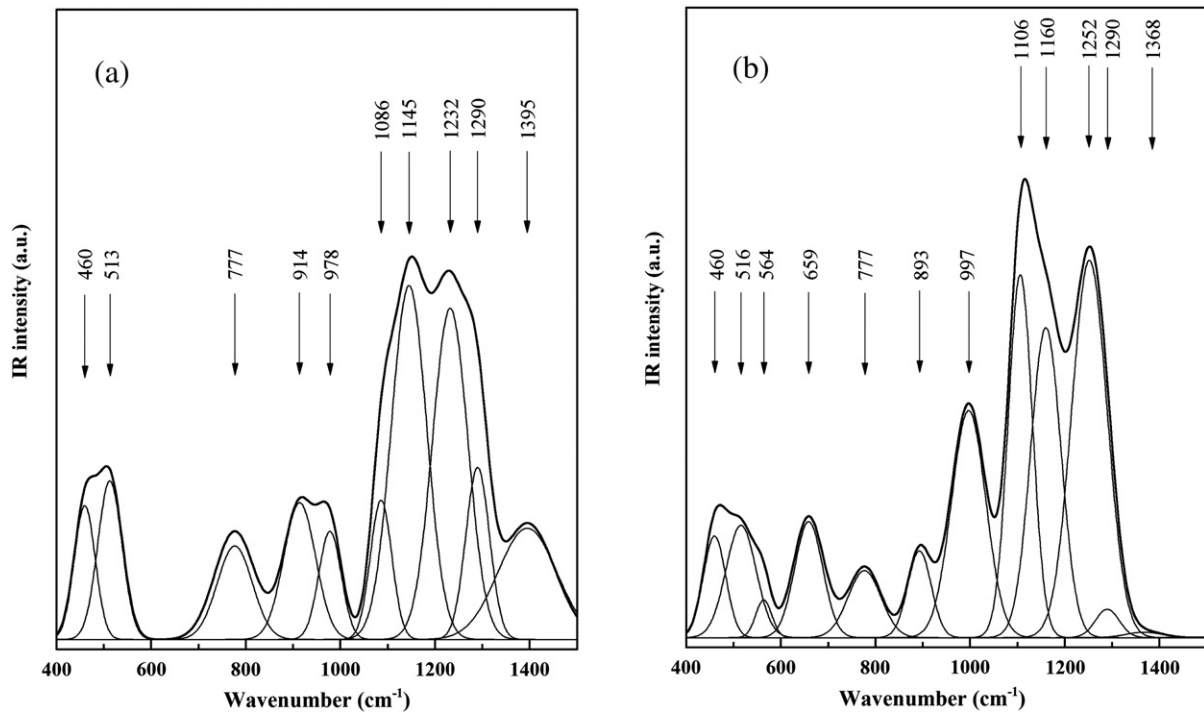


Fig. 9. Deconvoluted FTIR spectra of (a) $x = 0$ mol% and (b) $x = 5$ mol% of Fe_2O_3 containing oxyfluorophosphate glasses.

5 mol% of Fe_2O_3 in the ZCP glass indicates that the non-bridging oxygens in the glass network increase due to the replacement of less stable P–O–P bonds by chemically durable P–O–Fe bonds [60,68]. The increase in O–P–O linkages of $(\text{PO}_2)^-$ groups in Q^2 units which are responsible for the increase of P–O–Fe bonds can also be attributed to the disproportionation of Q^3 and Q^1 units (Fig. 7) as per the reaction [84], $2\text{Q}^2 \rightarrow \text{Q}^1 + \text{Q}^3$. The increase in the relative area (Table 5) of the band at 508 cm^{-1} due to Fe–O bonds supports the claim that the cleavage of P–O–P bonds and creation of P–O–Fe bonds. However, Raman spectra of the studied glasses do not account for separate Fe^{2+} and Fe^{3+} ions in general and octahedral FeO_6 and tetrahedral FeO_4 structural units in particular in the glasses. To throw more light on the structural modifications taking place in the glass regarding the presence of different phosphate and iron structural units, infrared spectra of the glasses is carried out as discussed below.

4.5. Infrared spectral studies

After deconvolution, four peaks each are observed in the absorption regions $1400\text{--}1120\text{ cm}^{-1}$ and $1120\text{--}750\text{ cm}^{-1}$ and two peaks in the

region below 750 cm^{-1} of pure glass (ZCP). The vibrational bands obtained in the high frequency region (i) at 1395 and 1290 cm^{-1} are attributed respectively to symmetric and asymmetric stretching vibrations of P=O bonds in Q^3 tetrahedra [54,86] and (ii) at 1232 and 1145 cm^{-1} are identified due to asymmetric and symmetric stretching vibrations respectively of $(\text{PO}_2)^-$ groups (NBOs) of Q^2 units [11,87]. The IR bands in the $1120\text{--}750\text{ cm}^{-1}$ region (i) at 1086 and 978 cm^{-1} are attributed respectively to $(\text{PO}_4)^{3-}$ groups of Q^0 phosphate units [54] and (ii) at 914 and 777 cm^{-1} are assigned to P–O–P linkages of bridging oxygens of Q^1 units [54] respectively. Finally, in the low wavenumber region, a band at 513 cm^{-1} due to network bending vibrations [54] and another band at 460 cm^{-1} due to ZnO_4 [1] tetrahedral units are evident in the deconvoluted IR spectrum of the pure glass. It may be noticed here that the vibrational band corresponding to asymmetric stretching vibrations of P=O bonds of Q^3 units is not Raman active in the present glass system.

Introduction of 0.5 mol% Fe_2O_3 into the glass matrix produced an additional peak at 567 cm^{-1} due to the vibrations of P–O–Fe bonds in FeO_6 octahedral units [79]. As Fe_2O_3 content is progressively increased from 0.5 to 1.5 mol%, (i) it is noticed that the wavenumber

Table 6

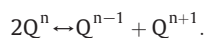
Wavenumbers (C) and relative areas (A) of deconvoluted bands in the IR spectrum of Fe_2O_3 doped oxyfluorophosphate glasses (accuracy of wavenumbers of band centers is $\pm 1\text{ cm}^{-1}$).

x (mol%)		$\nu_s(\text{P=O})$	$\nu_{as}(\text{P=O})$	$\nu_{as}(\text{O-P-O})$	$\nu_s(\text{O-P-O})$	$\nu_{as}(\text{P-O}^-)$	$\nu_s(\text{P-O}^-)$	$\nu_{as}(\text{P-O-P})$	$\nu_s(\text{P-O-P})$	FeO_4	FeO_6	$\delta(\text{O=P-O})$	ZnO_4
0	C	1395	1290	1232	1145	1086	978	914	777	–	–	513	460
	A	12	7.5	21.5	22.9	5.5	4.5	8	6	–	–	7	5.1
0.5	C	1391	1290	1228	1143	1088	980	920	777	–	567	517	460
	A	10.2	7	18.7	20.9	7.2	5.4	8.4	6.4	–	1	8.4	6.3
1	C	1388	1290	1225	1142	1091	981	923	777	–	572	523	460
	A	8	5.7	17.8	18.1	8.1	5.8	10.9	6.9	–	2.1	9.3	7.1
1.5	C	1384	1290	1220	1139	1093	984	925	777	–	581	528	460
	A	7.1	4.8	14.4	11.1	8.9	8.8	11.5	10.5	–	3.2	10.5	9.2
2	C	1381	1290	1225	1144	1097	986	917	777	633	578	525	460
	A	5	4.5	16.1	12.6	10.2	11.8	7.1	8	3.5	2.7	10.1	8.4
3	C	1377	1290	1233	1153	1099	989	906	777	640	574	522	460
	A	3.1	3.2	18.9	14.5	13.5	12.6	5.8	6.2	5.1	2	9.3	6
4	C	1371	1290	1244	1156	1102	994	899	777	648	570	518	460
	A	1.4	2.3	20.7	16.3	14.9	13.5	4.7	4.9	6	1.7	8.4	5.1
5	C	1368	1290	1252	1160	1106	997	893	777	659	564	516	460
	A	0.4	1.2	24.8	17.4	16.2	14.4	3.8	4.1	6	1.1	6.3	4.2

of the band at 460 cm^{-1} due to ZnO_4 and metal cation structural units is found to be constant with an increase in area and blue shift of deformation vibrational band at 513 cm^{-1} with increasing area, (ii) a decrease in both the wavenumbers and relative areas of the double bonded character $\text{P}=\text{O}$ (at 1395 and 1290 cm^{-1}) of Q^3 units and the non-bridging $(\text{PO}_2)^-$ structural groups of $\text{O}-\text{P}-\text{O}$ linkages of metaphosphate Q^2 units (at 1232 and 1145 cm^{-1}) is observed with a corresponding increase in the wavenumbers and relative areas of the pyrophosphate dimeric $(\text{P}_2\text{O}_7)^{4-}$ groups of Q^1 units (at 914 cm^{-1}) and the isolated orthophosphate $(\text{PO}_4)^{3-}$ groups of Q^0 units (at 1086 and 978 cm^{-1}). However, the wavenumber of the symmetrical stretching vibrations of $\text{P}-\text{O}-\text{P}$ linkages of Q^1 units at 777 cm^{-1} is found to be constant with an increase in relative area. The variations of the relative areas of different FTIR vibrational bands of Fe_2O_3 doped ZCP glasses with the concentration of Fe_2O_3 are depicted in Fig. 10. The increase in $\text{P}-\text{O}-\text{P}$ linkages (Fig. 10) with a simultaneous decrease in the non-bridging oxygen $\text{O}-\text{P}-\text{O}$ linkages lessens the connectivity of the glass network. Also, with the increase in Fe_2O_3 up to $1.5\text{ mol}\%$, the observed blue shift with an increase in relative area of the octahedral FeO_6 units at 567 cm^{-1} which plays the role of network modifier indicates the depolymerisation of the glassy network leading to a decrease in density and glass transition temperature.

With the further increase of Fe_2O_3 above $1.5\text{ mol}\%$, the diminishing of $\text{P}=\text{O}$ (Q^3) and $\text{P}-\text{O}-\text{P}$ stretching bands (Q^1) with simultaneous increase in the relative areas of $(\text{PO}_2)^-$ and $(\text{PO}_4)^{3-}$ stretching vibrational bands (Fig. 10) indicates the opening of $\text{P}=\text{O}$ bonds and breaking of $\text{P}-\text{O}-\text{P}$ linkages resulting in the enhancement of non-bridging oxygens through covalent $\text{P}-\text{O}-\text{Fe}$ linkages leading to length shortening of the phosphate chains which promotes the crosslinking between the shorter phosphate chains in the glass structure. This conversion of

various phosphate units in the glass network is in accordance with the disproportionation reaction [84]:



The decrease of Q^1 chain terminator also indicates that the glass network becomes more cross-linked with increasing Fe_2O_3 content [74]. At $2\text{ mol}\%$ of Fe_2O_3 , a new vibrational band is observed at 633 cm^{-1} which is assigned to $\text{P}-\text{O}-\text{Fe}$ linkages of Fe^{3+} ions in FeO_4 tetrahedral units [88]. With the further increase of Fe_2O_3 in the glass, the wavenumber and relative area of this band are observed to increase indicating the conversion of some of FeO_6 units into FeO_4 units which are responsible for the strengthening of the glass network.

In the present study, it may be noted that in addition to the variation of Fe_2O_3 to P_2O_5 ratio, the ZnO to P_2O_5 ratio and the CaF_2 to P_2O_5 ratio increase with increasing x . This can also affect the structure of the present glass and their chemical durability. The overall structure and hence the chemical durability is possibly due to a combination of varying ratios of the different modifier and former components (Fe_2O_3 , ZnO and CaF_2) to that of the glass forming oxide P_2O_5 and the cumulative effect of this can push the disproportionation reactions in either direction. In the case of phosphate glasses containing ZnO , it has been observed that a decrease in ZnO to P_2O_5 ratio increases the corrosion resistance of the glass [9] due to the decrease in Q^2 sites. The observed structural changes in the present glass system may be due to the presence of CaF_2 in the glass matrix.

Thus, the infrared spectral studies of Fe_2O_3 doped $\text{ZnO}-\text{CaF}_2-\text{P}_2\text{O}_5$ glasses indicate that $\text{P}-\text{O}-\text{P}$ bonds are believed to be replaced by $\text{P}-\text{O}-\text{Fe}^{2+}$ and $\text{P}-\text{O}-\text{Fe}^{3+}$ bonds. It has been reported that in iron phosphate glasses Fe(II) ions are in octahedral coordination, whereas Fe(III) ions have octahedral and tetrahedral coordinations [50,51]. The structural changes taking place in the glass matrix with the addition of Fe_2O_3 are well reflected in various properties studied like density, molar volume, glass transition temperature, thermal stability and relative areas of various vibrational bands.

5. Conclusions

The studies of Fe_2O_3 doped oxyfluorophosphate glasses revealed systematic changes corresponding to the structural linkages. FTIR and Raman spectral studies indicate the presence of characteristic phosphate structural units and the iron ions seem to cause a change in the short-range order structure of the glass matrix by replacing easily hydrated $\text{P}-\text{O}-\text{P}$ bonds by corrosion and moisture resistant highly stable $\text{P}-\text{O}-\text{Fe}$ bonds which enhance the cross-linking of the network. Up to $1.5\text{ mol}\%$, Fe_2O_3 in ZCP glass only serves to break down the extended polymeric network leading to the weakening of the glass network. However, above $1.5\text{ mol}\%$ of Fe_2O_3 in ZCP glass, enhanced cationic cross-linking seems to predominate leading to an overall strengthening of the glass network. The trend in the vibrational spectra matches well with the observed changes in the glass transition temperature, density and dissolution rate of these glasses. Finally, all these studies revealed that the stability of the glass is more when Fe_2O_3 content in the glass is above $1.5\text{ mol}\%$ and the glass with $5\text{ mol}\%$ of Fe_2O_3 is the most stable one.

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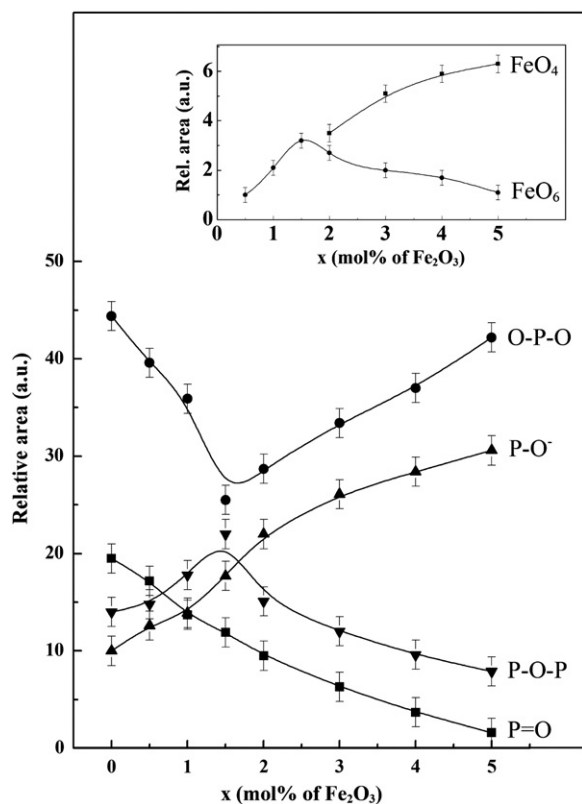


Fig. 10. Variation of the relative areas of IR bands of different phosphate units in ZCP glasses with the mol% of Fe_2O_3 (Inset represents relative areas of iron structural units & lines are drawn as a guide to the eye).

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